Cognitive State Recognition Using Deep Learning

Emily Lou[1,*], Mauro Victorio[2] and Dr. Vladimir Pozdin[3]

[1] Department of Computer Science and Engineering, Texas A&M University, College Station, TX

[2] Department of Electrical and Computing Engineering, Florida International University, Miami, FL [3]Department of Mechanical and Materials Engineering, Florida International University, Miami FL

* Email: emilylou8155@gmail.com

Motivation



Depiction of mentally exhausted doctor [1]

Mental fatigue monitoring is vital for jobs or tasks where high cognitive levels are imperative

A device placed on the forehead collects functional near-infrared spectroscopy (fNIRS) data to monitor cognitive activity using non-invasive techniques suitable for daily activities

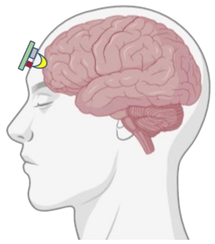


Image of device placed on forehead [2]

Methods

LSTM machine learning models are efficient for evaluating time-series data due to its ability to remember previous inputs. Compared to traditional recurrent neural networks, LSTM is unique in that it solves the issue of exploding or vanishing gradients due to its hidden state propagation

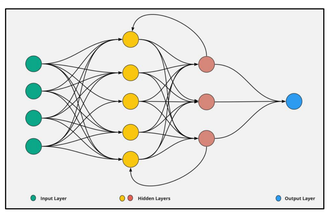


Figure 1. Recurrent neural network architecture [3]

The LSTM model was trained on a dataset consisting of the change in oxygenated hemoglobin (HbO) and change in deoxygenated hemoglobin (Hbb) collected from 10 participants using the initial prototype of the wearable device that is not flexible.

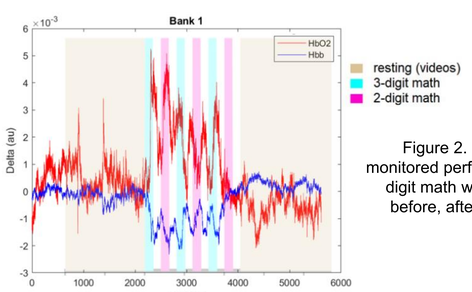


Figure 2. Participants are monitored performing 2-digit and 3-digit math with resting periods before, after, and in between

Binary classification of active vs resting periods as well as multi-classification of hard math, moderate math, and resting periods were evaluated. Various model architectures, pre-processing methods, additional features, and training techniques were tested.

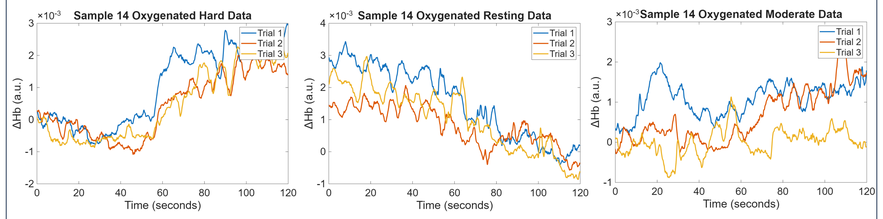


Figure 2. Change in oxygenated and deoxygenated hemoglobin during various levels of cognitive activity across one user sample

Pre-Processing - Each sample was first smoothed with a moving average of 10. Signals were then normalized by either per-sample or across samples at each time step using z-score scaling then per-sample baseline correction

Segmentation - Each 120 second sample was split into 45-second intervals with a 5 second overlap

Feature Extraction - The total change in hemoglobin was computed for each sample before normalization. The first and second derivatives of both HbO and Hbb were extracted for each segment after normalization.

Training & Validation - A group 4-fold cross validation was utilized during model training

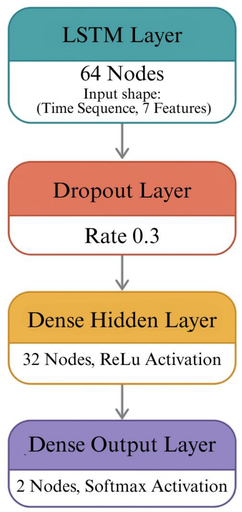


Figure 3. Model Architecture [4]

Conclusions

- The LSTM model achieved an accuracy of ~65% for classifying the active and resting cognitive data, demonstrating its potential for interpreting temporal neural signals
- The model performed well classifying high and resting levels of cognitive activity but had difficulty classifying moderate levels. The model exhibited signs of overfitting and poor generalization due to the gap between training and validation accuracy, likely due to the small dataset size
- Incorporating z-score normalization, segmenting the time-series data using overlapping windows, and the total change in hemoglobin feature, significantly enhanced model performance.
- Although z-score normalization per sample is the standard, z-score normalization across samples at each time-step yielded higher metrics

Acknowledgements

This research was supported by the National Science Foundation (NSF) under award 2349464.

Results



Figure 4. Training and validation accuracy and loss for model 1. An increase in training accuracy and decrease in training loss indicates learning. However, a fluctuating validation accuracy and loss indicates poor generalization and possible overfitting

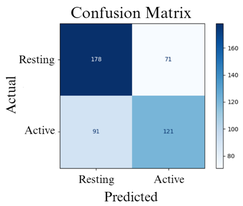


Figure 5. Confusion matrix for model 1. Model does well classifying resting samples, while struggling with active samples

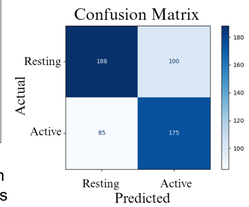


Figure 6. Confusion matrix for model 3. Model does approximately equally well classifying resting, moderate, and hard data

Model	Classification Type	Z-score Normalization	Accuracy	Precision	Recall	F1 Score
1	**Active, Resting**	**Per Sample**	**0.6486**	**0.6472**	**0.6486**	**0.6379**
2	Hard, Moderate, Resting	Per Sample	0.4638	0.4671	0.4638	0.4516
3	Active, Resting	Across Samples	0.6624	0.6638	0.6624	0.6626
4	Hard, Moderate, Resting	Across Samples	0.5474	0.5545	0.5474	0.5485

Figure 6. Comparison of LSTM models with accuracy, precision, recall, and F1 score metrics

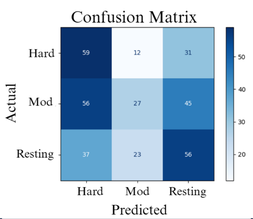


Figure 7. Confusion matrix for model 2. Model does well classifying hard and resting samples, but struggles with moderate samples

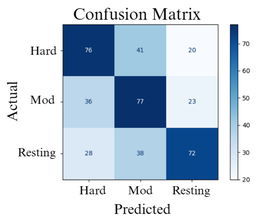


Figure 8. Confusion matrix for model 4. Model does approximately equally well classifying resting, moderate, and hard samples

Ongoing Works

The LSTM model will be evaluated on fNIRS data collected from the flexible, wearable version of the device, which is expected to produce higher quality signals due to closer contact with the forehead. Additionally, a dataset with a greater number of participants will likely enhance model generalizability and reduce overfitting. Future efforts will also explore other model architects or features such as additional layers or attention

References

[1] Image from Iconscout.com [2] Image designed with BioRender [3] Image from dataspraint.com [4] Khan, M. A., Asadi, H., Qazani, M. R. C., Arogbonlo, A., Pedrammehr, S., Anwar, A., Zhou, H., Wei, L., Bhatti, A., Oladazimi, S., Khan, B., & Nahavandi, S. (2025). Enhancing cognitive workload classification using integrated LSTM layers and CNNs for fNIRS data analysis. Computers, 14(2), 73 [4] Image generated by Perplexity [5] Staudemeyer, R. C., & Morris, E. R. (2019, September 23). Understanding LSTM: A tutorial into Long Short-Term Memory Recurrent Neural Networks. Schmalkalden University of Applied Sciences & Singapore University of Technology and Design.